

PROJECT REPORT

Dynamic QR Asset Tracker

Python · SharePoint · Microsoft Lists · Power Automate · Excel

Role	Analyst & Automation Engineer - End to End
Domain	IT Asset Management
Tech Stack	Python, qrcode, Pillow, pandas, openpyxl, SharePoint
Scale	496+ IT Assets tracked organisation-wide
Output	Print-ready A4 PDF sticker sheets at 300 DPI

1. PROJECT NEED & BACKGROUND

Problem Statement

The organisation manages hundreds of IT assets across multiple office locations. Before this project, the IT team had no reliable, real-time method to identify an asset, its owner, warranty status, or configuration — without manually searching through Excel sheets.

Pain Points

- **No Physical Identification:** Laptops had no labels — field staff could not identify assets without opening spreadsheets.
- **Manual Tracking Risk:** Assets were tracked only in Excel — prone to human error, version mismatch, and data loss.
- **Warranty Blindspot:** Expiry dates not visible at a glance — IT team missed warranty renewals.
- **Slow Audit Process:** Each asset had to be manually cross-referenced — taking hours instead of seconds.
- **No Org-wide Visibility:** Departments had no self-service way to view asset details without contacting IT.

Solution

Build a Python automation that reads asset data from a master Excel file, generates a unique QR code per asset (linked to the SharePoint record), prints it as a professional sticker with asset ID and company logo, and outputs a print-ready PDF. Stick QR labels on every asset - any staff member can scan with a phone to instantly see full live asset details on SharePoint.

2. PROJECT WORKFLOW - End to End

STEP 1 - Excel Input	Master asset list with 496+ rows, 19 columns: Asset ID, Brand, Serial No., Warranty, ItemLink to SharePoint record.
STEP 2 - BAT File Trigger	IT admin double-clicks run_qr_generator.bat - no coding needed. Calls the Python script automatically.
STEP 3 - SharePoint URL	For each row, script reads ItemLink column or auto-builds the SharePoint DispForm URL using asset ID.
STEP 4 - QR Generated	qrcode library generates ERROR_CORRECT_H QR (30% damage tolerance). Company logo embedded at centre.
STEP 5 - Sticker Designed	Pillow composes sticker: QR left, Asset ID + logo right, all within a rounded border at 300 DPI.
STEP 6 - PDF Generated	18 stickers per A4 page (3x6 grid). All pages saved as single print_pages.pdf at 300 DPI.
STEP 7 - Print & Stick	IT team prints sticker sheets and attaches labels to every asset.

STEP 8 - Live Lookup

Employee scans QR -> opens SharePoint record showing assignee, specs, warranty status.

3. TECHNOLOGY STACK - Libraries & Packages

pandas

Install	pip install pandas
Category	Data Processing
What it is	Industry-standard Python library for data manipulation. Works with Excel, CSV, SQL.
Why we used it	Read the 496-row Excel file efficiently, access columns by name, handle NaN checks.
Key APIs	<code>pd.read_excel(path, engine='openpyxl')</code> <code>df.iterrows()</code> <code>pd.isna(value)</code>

qrcode

Install	pip install qrcode[pil]
Category	QR Code Generation
What it is	Python library to generate QR codes with error correction levels and PIL output.
Why we used it	ERROR_CORRECT_H (30% damage tolerance) critical because logo covers ~20% of QR centre.
Key APIs	<code>qrcode.QRCode(error_correction=ERROR_CORRECT_H)</code> <code>qr.add_data(url)</code> <code>qr.make_image()</code>

Pillow (PIL)

Install	pip install Pillow
Category	Image Processing & Sticker Design
What it is	Most widely used Python imaging library. Handles image creation, drawing, compositing.
Why we used it	Create canvases, draw text, paste logos with transparency, draw rounded borders, export PDF.
Key APIs	<code>Image.new()</code> <code>draw.text()</code> <code>draw.rounded_rectangle()</code> <code>image.paste(logo, pos, mask)</code> <code>image.save(..., 'PDF', resolution=300)</code>

openpyxl | os | logging | typing

openpyxl	Excel engine used by pandas to read .xlsx files. Specified as <code>engine='openpyxl'</code> in <code>pd.read_excel()</code> .
os	<code>os.makedirs()</code> creates output folders; <code>os.path.exists()</code> implements skip logic for existing stickers.
logging	Timestamped console output so operator can track progress across 496 assets.

<code>typing.Optional</code>	Type hint <code>Optional[str]</code> means <code>str</code> or <code>None</code> . Makes function signatures readable and IDE-friendly.
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4. CODE WALKTHROUGH & KEY EXPRESSIONS

4.1 QR Code Generation with Logo

```
def generate_qr_image(data, logo_path=None, ...):
    qr = qrcode.QRCode(error_correction=qrcode.constants.ERROR_CORRECT_H)
    qr.add_data(data)          # data = SharePoint URL
    qr.make(fit=True)
    qr_img = qr.make_image(fill_color='black', back_color='white').convert('RGBA')
    logo = Image.open(logo_path).convert('RGBA')
    logo_size = int(qr_w * 0.20) # logo = 20% of QR width
    pos = ((qr_w - lx) // 2, (qr_h - ly) // 2) # center
    qr_img.paste(logo, pos, mask=logo)          # transparent overlay
```

4.2 Sticker Composition

```
sticker = Image.new('RGB', (sticker_w, sticker_h), 'white')
draw = ImageDraw.Draw(sticker)
sticker.paste(qr_img, (qr_x, qr_y)) # QR on left
draw.text((text_x, text_y), 'Asset ID: ' + asset_id) # text on right
sticker.paste(right_logo, (logo_x, logo_y), mask=logo) # logo below text
draw.rounded_rectangle([...], radius=24, outline='#000000', width=6)
```

4.3 A4 Grid Assembly & PDF Export

```
col = pos_in_page % COLS # column 0,1,2
rowpos = pos_in_page // COLS # row 0..5
paste_x = PAGE_MARGIN + col * (sticker_w + GUTTER_X)
paste_y = PAGE_MARGIN + rowpos * (sticker_h + GUTTER_Y)
page_images[0].save(PDF_OUTPUT, 'PDF', resolution=300, save_all=True,
append_images=page_images[1:])
```

5. BUSINESS BENEFITS & IMPACT

Metric	Before	After
Time to identify an asset	5-10 min (search + call)	< 5 seconds (QR scan)
Audit duration	2-3 days manual	Half a day
Sticker creation	N/A (none existed)	1 BAT click, automated
Warranty miss risk	High (manual)	Zero (SharePoint alerts)
Assets tracked	Excel only	496+ live on SharePoint